Decoding Customer Value: A SQL-Driven Retention Strategy

Background

A direct-to-consumer (D2C) fashion brand sells clothing, accessories, footwear, and outerwear across the United States. The brand has no physical stores and no third-party retailers — every customer relationship is managed directly by the brand itself.

The brand has grown steadily and now has customer behavioral data covering around 3,900 customers. It runs a promotional discount program, supports multiple payment methods, and offers a range of shipping options. But it has never built a structured way to understand its customers beyond surface-level sales numbers.

The founding team is at a critical point: they can keep running promotions reactively and hope customers keep coming back — or they can build something more deliberate. They have chosen to build.

Core Problem

The brand has data but no intelligence built on top of it. Specifically, it cannot answer:

- Who are the customers likely to still be buying two years from now, and what do they look like today?
- Is the discount and promo program actually building a loyal customer base, or just attracting one-time bargain hunters?
- Which product categories are associated with lower purchase history, and which ones appear most among high-frequency, high-tenure customers?
- Are there cities or regions where the brand has strong traction that it has not yet deliberately targeted, and does that traction look different in terms of category preference, promo sensitivity, or average spend per customer?
- What does the brand's best customer actually look like in terms of age, purchase habits, payment preferences, and satisfaction?

Without answers to these questions, the brand is making marketing and product decisions based on gut feel rather than evidence.

Problem Statement

Using only transactional and behavioral data, can the brand identify what its most valuable customers look like, measure how much of its current revenue depends on promotions, and build a data-backed retention strategy that reduces discount dependency without hurting sales?

Key Questions to Address

1. Who are the genuinely loyal customers vs. those who only buy when there is a discount?
2. What behavioral patterns today predict high customer value over time?
3. Which geographies and demographics are commercially underlevered?
4. How should the brand restructure its promotional strategy to protect margins without losing volume?
5. What does the brand's ideal customer profile look like, and how can it acquire more of them?

Scope of Analysis

1) Data Preparation & Feature Engineering (Python):

Clean and prepare the raw dataset. Then build the customer-level metrics the brand needs — but here is the constraint: you are not told which metrics to build. You need to decide what to measure and justify why those choices make sense for this business.
As a starting point, think about what signals in the data could indicate value, satisfaction, and promotional reliance. But do not stop at computing numbers, explain the logic behind each metric you create and what it is actually trying to capture.
One thing to keep in mind: metrics that sound analytical but do not lead to a decision are not useful. Every engineered feature should answer a question the brand actually cares about.

2) Customer Segmentation & Analysis (SQL)

Build a structured query layer to answer the brand's core business questions:

- What separates high-value customers from low-value ones, and which profiles show the strongest repeat purchase behavior?
- Which seasons and categories are associated with lower-tenure customers versus those with high previous purchase counts?
- Which geographies signal organic demand versus discount-driven volume?

3) Founder Dashboard (Power BI)

Build a clean, four-panel dashboard designed for a non-technical founding team:

- Customer pyramid: showing how value is distributed across the customer base
- Promo dependency vs. retention rate: plotted by segment to show who needs discounts to buy and who doesn't
- Geographic opportunity map: not just which regions buy most, but which regions show high spend and low promo dependency, indicating genuine brand pull rather than discount-driven demand
- Category funnel: showing which product categories are associated with low purchase history versus high purchase history, as a proxy for entry-point versus retention categories

4) Retention Playbook (Business Recommendations)

Translate your findings into two outputs the brand can act on. Every recommendation must state the trade-off clearly — not just what to do, but what you risk by doing it.
Promotional sunset plan: Identify which segments to gradually stop discounting, why those segments specifically, and what the margin impact looks like. The recommendation must name the segment, the trigger behavior, the rollout timeline, and the metric you would track to know if it is working. "Reduce discounts" is not an acceptable answer.
Ideal customer profile: A data-backed description of the brand's most valuable customer type, specific enough that a marketing team could use it to make targeting decisions today.

Dataset

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The Central Analytical Challenge

The Core Constraint

The dataset has no loyalty score, no churn label, and no timestamps. Every concept you use must be constructed from available variables, not assumed.

Loyalty must be defined, not declared. You are required to build at least two competing definitions using different variable combinations, test both, and argue clearly for one. Your justification must be grounded in internal consistency, correlation with revenue, or predictive logic — not intuition.

Every segment claim must be traceable. Labels like "at-risk" or "high-value" are only valid if they map to a specific, stated combination of variables. Assertions that cannot be traced back to the data will not meet the standard of this project.

Expected Outcome

A comprehensive customer intelligence report and interactive dashboard designed to provide the brand with a clear and actionable answer to a critical strategic question:

"Is the business successfully building a loyal customer base, or is it reliant on continuous promotional activity? Furthermore, what strategic actions should be taken under either scenario?"

Deliverables

Python	Cleaned dataset + engineered features (dependency score, value tier, satisfaction flag)
SQL	Segmentation queries answering all 5 key questions above
Power BI	Four-panel founder dashboard as described in scope
Playbook	Promo sunset plan + ideal customer profile
Summary	A concise executive summary (max 1 page) of findings and recommendations

Point of Contact

Achyuth - 6381774762
Dhairya Nisar - 8928149400

Resources

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